Tanmay Bhuskute

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SUMMARY

DevOps Engineer with 2+ years running production Kubernetes migrations and CI/CD pipelines at scale (**40+ environments**, **50% downtime reduction**) on AWS with Docker, Terraform, Ansible, and GitLab CI/CD; now extending that infrastructure-automation experience into applied research as an M.S. Computer Science candidate (Dec 2026).

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, ArgoCD, Helm, Terraform, Ansible.

CI/CD & Automation: GitLab CI/CD, Jenkins, Git, Bash, Python scripting, Nginx, HashiCorp Vault.

Databases & Caching: PostgreSQL, MySQL, Redis, Elasticsearch.

Networking & Reliability: Linux, TCP/IP, DNS, SSL, VPC, SonarQube, HA/Backup design.

Programming: Java, Python, JavaScript, TypeScript, Go, SQL.

WORK EXPERIENCE

Graduate Researcher — University of Cincinnati Feb 2026 – Present

- Built config-driven experimental automation harness with device-orchestration and stall detection, evaluated across 53 runs on 20+ Android apps, achieving **25.7x latency reduction** over baseline.

Software Development Engineer — Siemens Industry Software Jul 2022 – Jun 2024

- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault, modernizing cloud-native deployment architecture.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with automated rollback and recovery.
- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern — Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern — Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, PostgreSQL, Redis, Docker] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset); served via containerized FastAPI backend with PostgreSQL and Redis caching, deployed on Oracle Cloud.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Docker, MCP] May 2026

- Built full-stack TypeScript app (React SPA + Express 5 REST API + MCP server) with Zod schema validation at every data boundary, eliminating a class of runtime type-mismatch bugs across all external tool calls.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

CERTIFICATIONS & ACTIVITIES

- **SAFe Practitioner 5.1** (Mar 2023).
- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools).